

Multiple Choice

1. **Answer:** A

Options: A) -.27, 1.45, 1.73; B) -.27, -1.45, -1.73; C) -2.18, 0.15, -1.82; D) 0, 1.5, -2; E) .27, 1.45, 1.73

Note: The correct Z-scores for the raw scores of 115, 134, and 99 are calculated using the formula $Z = (X - \mu) / \sigma$, where X is the raw score, μ is the mean (118), and σ is the standard deviation (11), resulting in Z-scores of approximately -0.27, 1.45, and -1.73, respectively.

2. **Answer:** B

Options: A) Savanna; B) Taiga; C) Tundra; D) Chaparral; E) Alpine

Note: The taiga, also known as the boreal forest, is characterized by its cold climate and the presence of permafrost, which limits the depth of plant roots, making it difficult for many species to grow deep-rooted systems.

3. **Answer:** A

Options: A) reproduction, migration, mutation, natural selection; B) mutation, genetic drift, migration, genetic hitchhiking; C) mutation, genetic drift, migration, random mating; D) mutation, gene flow, migration, natural selection; E) adaptation, genetic drift, migration, natural selection

Note: The major forces of evolution-reproduction, migration, mutation, and natural selection-are fundamental mechanisms that drive genetic variation, adaptation, and the evolution of species over time.

4. **Answer:** C

Options: A) Aristotle, Darwin, and Lamarck; B) Aristotle, Lyell, and Darwin; C) Aristotle, Linnaeus, and Cuvier; D) Aristotle, Cuvier, and Lamarck; E) Linnaeus, Darwin, and Lamarck

Note: Aristotle, Linnaeus, and Cuvier all held views that supported the idea of species being immutable and unchanging, reflecting a fixed hierarchy of life that was prevalent in their respective eras.

5. **Answer:** A

Options: A) photosynthesis to produce glucose; B) fermentation to produce alcohol; C) association with humans to produce ringworm; D) association with the roots of plants to form mycorrhizae

Note: Fungi do not perform photosynthesis to produce glucose because they are heterotrophic organisms that obtain their nutrients by decomposing organic matter rather than synthesizing their own food through photosynthesis like plants do.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Gene flow between populations tends to reduce genetic divergence and homogenizes genetic variation, which decreases the likelihood of speciation rather than increasing it.

7. **Answer:** False

Options: A) True; B) False

Note: Maintaining the mosquito population at the highest point of its logistic curve would lead to resource depletion and increased competition, ultimately destabilizing the population rather than ensuring long-term stability.

8. **Answer:** True

Options: A) True; B) False

Note: Research has shown that the presence of an extra Y chromosome in males, as seen in conditions like XYY syndrome, has been associated with lower levels of aggression and impulsivity compared to typical XY males, suggesting that the additional Y chromosome may influence behavioral traits.

9. **Answer:** False

Options: A) True; B) False

Note: Commensalism is indeed an interspecies interaction because it involves a relationship between different species, where one species benefits while the other is neither helped nor harmed.

10. **Answer:** True

Options: A) True; B) False

Note: The conversion of pyruvic acid to acetyl CoA involves decarboxylation and the addition of Coenzyme A, rather than the breakdown of a compound through the addition of water, which is characteristic of hydrolysis.

Long Form Response

11. **Answer:** The age of rock formations can be estimated through various methods, primarily utilizing radioactive elements and fossil content. Radioactive dating, specifically radiometric dating, measures the decay of isotopes within minerals, allowing geologists to determine the age of a rock based on the ratio of parent to daughter isotopes. Additionally, the fossil content found within rock layers plays a critical role in dating, as the presence of certain fossils, known as index fossils, can indicate specific time periods in Earth's history. By correlating the fossil assemblages with established geological time scales, scientists can further refine the age estimates of rock formations. Together, these methods provide a comprehensive approach to understanding the chronological sequence of geological events.

Note: correct because it accurately describes two primary methods for estimating rock

ages: radiometric dating, which relies on the decay of radioactive isotopes, and the use of index fossils to provide relative dating based on the presence of specific fossilized organisms within the rock layers.

12. **Answer:** Incomplete dominance and epistasis are both important concepts in genetics that illustrate different interactions between alleles and genes. Incomplete dominance occurs when two alleles of a single gene interact to produce a phenotype that is a blend of the traits associated with each allele, exemplified by the pink flowers resulting from the cross of red and white flowering plants. In contrast, epistasis involves the interaction between two different genes, where the presence of one gene's alleles can mask or modify the expression of another gene's alleles, leading to variations in phenotypic outcomes. While both concepts demonstrate complex genetic interactions, they differ fundamentally in that incomplete dominance focuses on the interaction within a single gene, whereas epistasis emphasizes the relationship between multiple genes. Understanding these concepts enhances our comprehension of genetic variation and inheritance patterns.

Note: The answer correctly distinguishes between incomplete dominance, which involves a blending of traits from two alleles of the same gene, and epistasis, which describes the interaction between alleles of different genes that can affect the expression of phenotypes.